Constant Current Linear Regulator with Active Power Factor Correction

Features

- 700V High Voltage Technology for High Reliability with 440 Vac Operation and 4kV Surge
- Closed-loop Control for Constant Output Current, and Excellent Line and Load Regulation to Ensure Constant Output Average Current within A Certain Input Voltage Range
- Real Time Line Voltage Monitoring and Power Regulation for Achieving An Effect of Constant Input Power
- Capacitance Amplifier Technology is Applied for Loop Compensation of Control with No MLCC
- A Multiplier is Designed for Very High Power Factor (>0.95) and Very Low THD (<15%) to Meet the Highest IEC Standard
- Current Foldback At Both Low Line and High Line for Thermal and Reliability Performance
- High Saturation Current, Supports upto 12W Applications, like bulb, Panel and Down Light
- Support Parallel Application with Multiple ICs for Higher Power Applications like Flood Light
- ESOP8 Package

Protection

- Intelligent Thermal Regulation
- Built-In Soft Start

Application

- Linear LED Driver with LED1:LED2:LED3=7:5:3, 18V LED chip

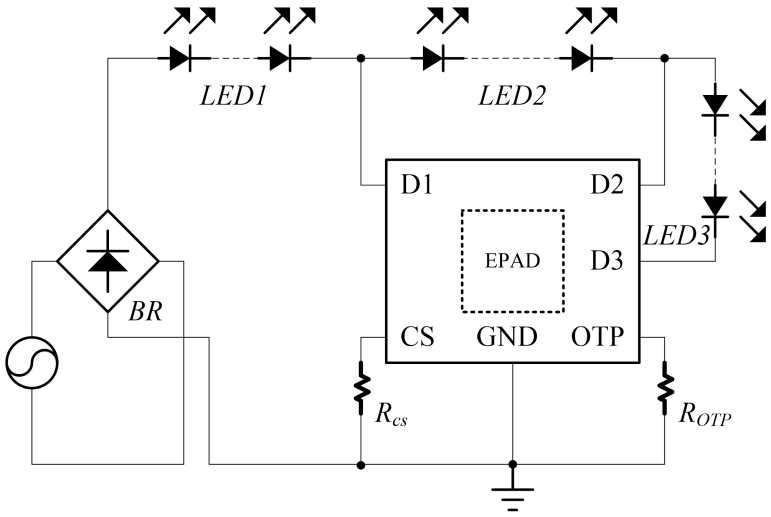


Fig. 1. Typical application

Wuxi Grandemicro Technology Co., Ltd. (hereinafter referred to as the "Grandemicro") reserves the right to change the product and service at any time without notice. Users are requested to obtain the latest information before purchasing the product and verify that the information is the best and complete. All products in the order confirmation will follow the regulations of Grandemicro. The contents of this information is strictly prohibited by other purposes to be reproduced or copied without the permission of Grandemicro. Loss on the use of this product without consulting with the sales department, Grandemicro does not assume its responsibility.

Address: Room 1501, Block A3, West Jianzhu Rd. 777, Binhu District, Wuxi, Jiangsu , China
Phone/Fax: 86-510-85388846

Package and Ordering Information

Part No.	Material Type	Package	Operating Temperature	Packing Method
YT2173E8	Green	ESOP8	-40 ℃ to 105 ℃	Tape 4000 pcs/Reel

Pin Configuration

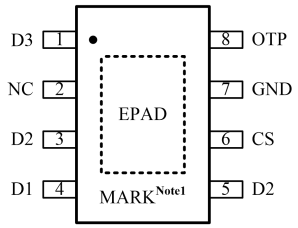


Fig. 2. Pin configuration

Note 1： Please refer to the "Marking Rule".

Pin Definition

Pin No.	Name	Description
1	D3	Drain of the Channel 3. Connect to The Cathode of The 3rd LED String.
2	NC	No Connection.
3, 5	D2	Drain of the Channel 2. Connect to The Cathode of The 2nd LED String.
4	D1	Drain of the Channel 1. Connect to The Cathode of The 1st LED String.
6	CS	Current Sense. Connect A Resistor to GND to Program The LED Current.
7	GND	Ground.
8	OTP	Over Temperature Protection Setting.
EPAD	NC	No Connection. Can be connected to GND for thermal purpose.

Internal Block Diagram

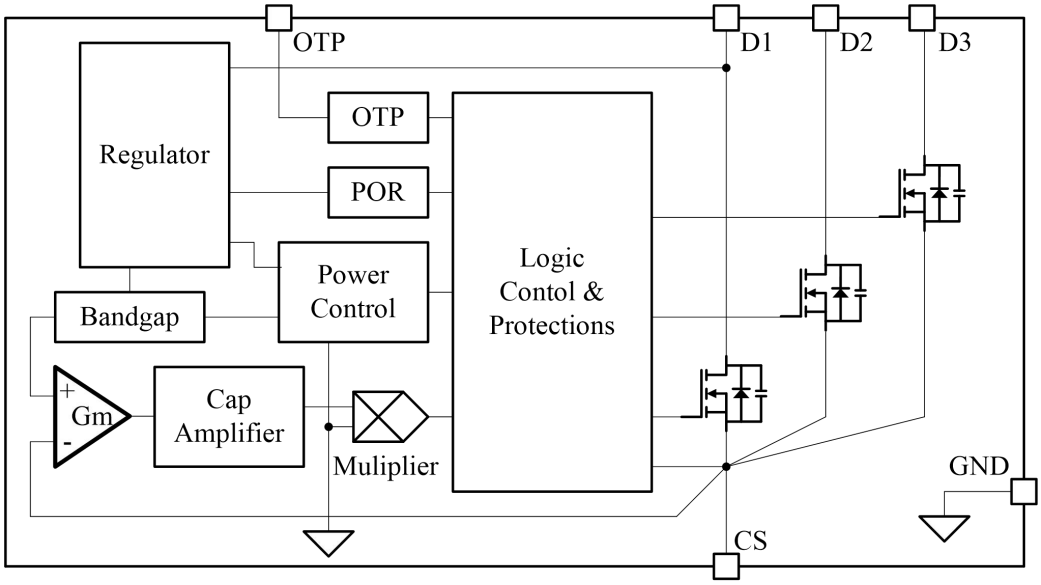


Fig. 3 Internal block diagram

Absolute Maximum Ratings[Note2]

Parameters	Symbol	Range	Units
"D1", "D2" and "D3" voltage range	V_{D1}, V_{D2}, V_{D3}	-0.3 ~ 700	V
"CS", "OTP" voltage range	V_{CS}, V_{OTP}	-0.3 ~ 5.5	V
ESD Human mode[Note3]	ESD_{hbm}	2000	V
Operating junction temperature range	T_j	-40 ~ 150	℃
Ambient temperature range	T_a	-40 ~ 105	℃
Storage temperature range	T_{stg}	-40 ~ 150	℃
Welding temperature (< 20 *s* welding)	T_{lead}	260	℃
Maximum dissipated power	P_{d_max}	2	W
Junction to ambient thermal resistance	θ_{thja}	50	℃/W

Note2: Stresses beyond those listed under "absolute maximum ratings" may cause permanent damage to the device.

Note3: Electrical components and circuit boards will be aware of the situation in the discharge is not easy. Although this product has a special electrostatic protection circuit, but in the case of high-energy electrostatic discharge, the chip may have damage and loss of function or performance degradation. Therefore, users still need to take appropriate preventive measures ESD.

Electrical Characteristics[Note4] (Unless otherwise specified, T_a = 25 ℃)

Parameter	Symbol	Conditions	Min	Typ.	Max	Units
Operation Current						
Operation Current	I_{OP}			160	210	μA
MOSFET						
BV of D1	V_{BV1}	I_d = 250μA	700			V
BV of D2	V_{BV2}	I_d = 250μA	700			V
BV of D3	V_{BV3}	I_d = 250μA	700			V
Saturation of D1	I_{SAT1}		55			mA
Saturation of D2	I_{SAT2}		95			mA
Saturation of D3	I_{SAT3}		125			mA
CS						
Internal Reference Voltage	V_{REF}		450	465	480	mV
OTP						
Thermal Regulation Threshold1	T_{OTR1}	OTP floating		145		℃
Thermal Regulation Threshold2	T_{OTR2}	OTP short to GND		135		℃
Thermal Regulation Threshold3	T_{OTR3}	OTP connect 100kohm resistor to GND		120		℃
OTP Slope	K_T			-2.3%		℃

Note4： The max and minimum parameters specified are guaranteed by test, the typical value are guaranteed by design, characterization and statistical analysis.

Application Suggestion

YT2173 is a 3-stages linear high-voltage LED driver for direct line operation, target at ultra wide (90-440Vac) withstanding line voltage range, high surge capability (4kV) and low THD requirement.

Theory of Operation

The input voltage (V_{in}) is the rectified voltage from AC lines by the bridge rectifier. When V_{in} is higher than the forward voltage of the 1st LED string, the chip can detect it and turn on the 1st N-MOS to lighten the 1st LED string. When V_{in} keeps increasing and exceeds the total forward voltage of the 1st and the 2nd LED strings, the chip turns on the 2nd N-MOS to lighten the 1st and the 2nd LED strings. In the same way, all LED strings are lightened as V_{in} increases.

During V_{in} decrease period, the chip shuts down the N-MOS as the reverse sequence.

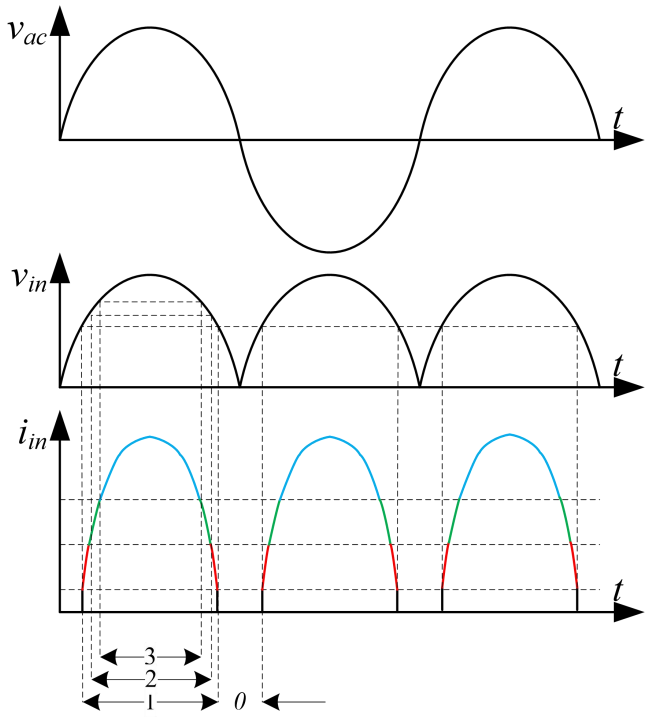


Fig. 4 Theory of operation and the key waveform

Sine Wave Current Shaping

The IC creatively introduces an analog multiplier into the linear IC to modulate the output LED current waveform, so that the output LED current has a sinusoidal waveform to achieve high PF and ultra-low THD, which can meet the IEC6100-3-2 harmonic standard. The input voltage and current waveform are shown as Fig. 4.

Input Power Control

The IC creatively introduces the closed-loop current control and adopts capacitor amplifier technology to eliminate the COMP capacitor for loop compensation. The average output current and the input power is set by the resistor R_{cs} between the CS pin and the chip ground. The average input power can be calculated with formula (1) where "0.86" is the efficiency coefficient and "20" is the amplifier coefficient.

$$P_{IN} = \frac{(20 \times V_{ref})^2}{0.86 \times R_{cs}} \quad (1)$$

Over Temperature Regulation

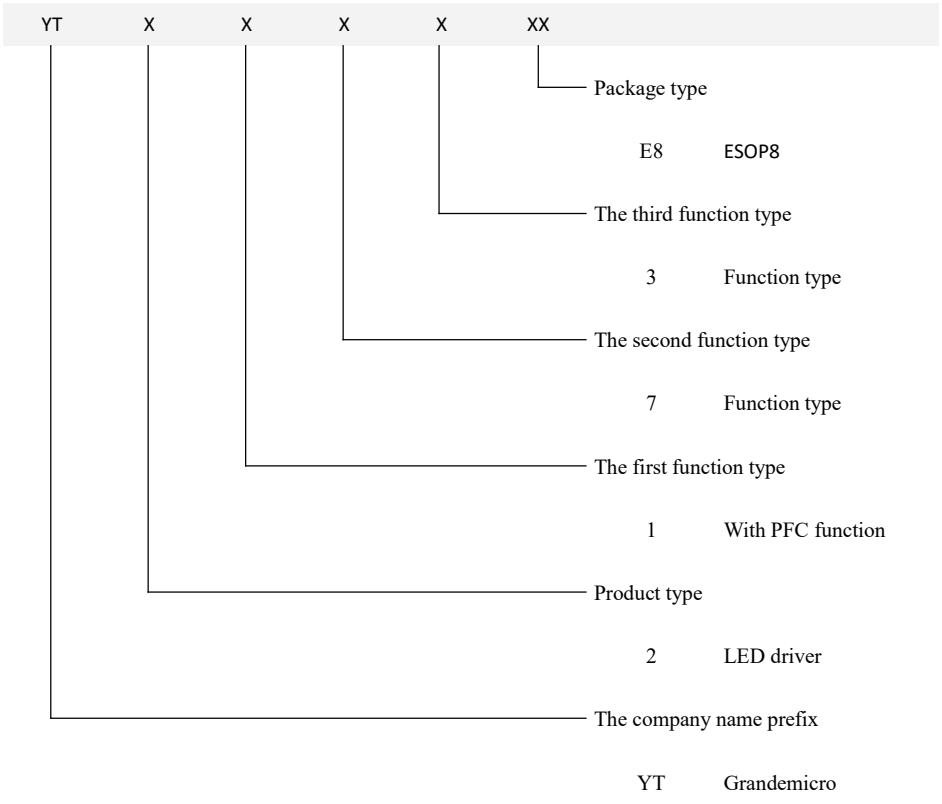
The IC integrates an intelligent over-temperature regulation. When the internal junction temperature of the chip is higher than T_{OTR}, the internal reference voltage decreases, and the output current decreases accordingly.

PCB Layout Suggestion

Use the trace for current sense resistor as short as possible.

The IC adopts ESOP8-EP package to strengthen the thermal dissipation, user can connect E-pad to GND to decrease thermal resistance.

Marking Rule



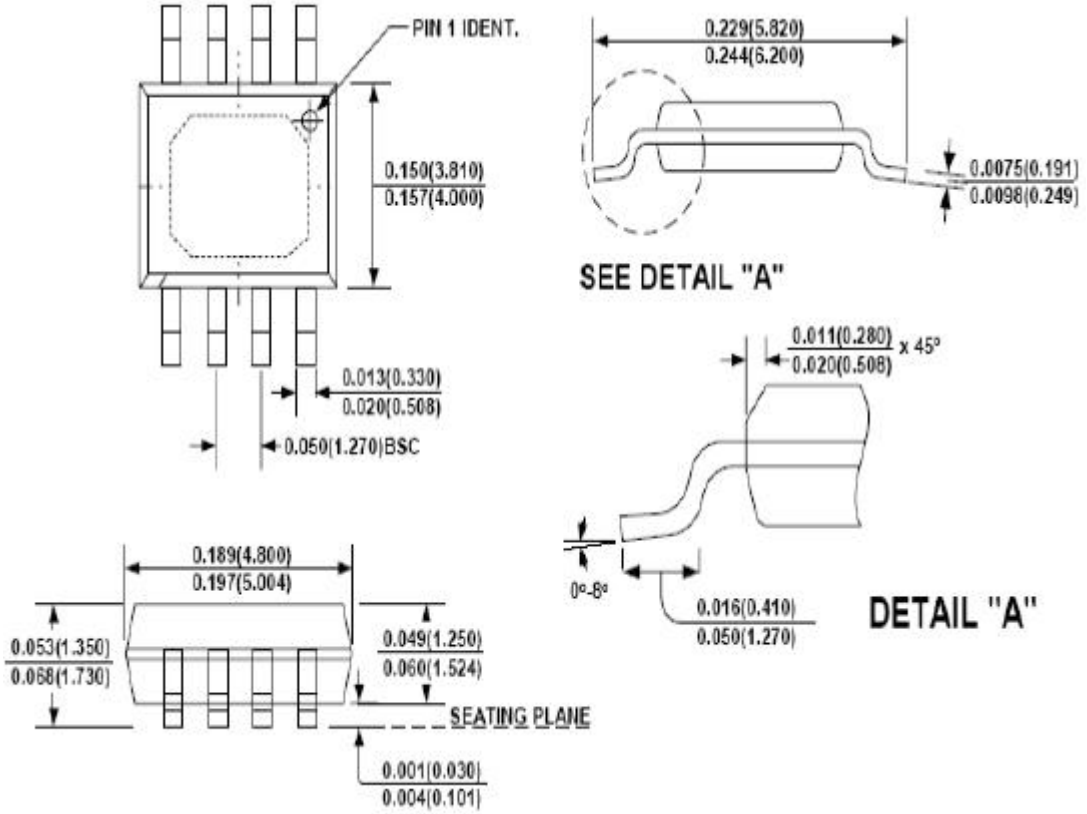
Part No.

●	P	Y	M	D	S	V	F

Part No.	YT2173E8
●	The first pin marked
Date code	P: Packing site; Y: Year; M: Month; D: Day; S: Series; V: Version; F: MOSFET.

Package

ESOP8



Version History

Version	Date	Description
A0	Dec. 2021	Draft
A1	Apr. 2023	Released
A2	May 2023	Update the pin map information
A3	Feb. 2024	Revised some wording
A4	Oct. 2024	Add maximum dissipated power information